EEEE4121

AC Drives Coursework 1

Simulation of the control and performance of an Induction motor with PLECS

AUTHOR:	**Mr Royalty Chihava**
ID NUMBER:	**20521383**
CW CONVENOR:	**Dr. Christian Klumpner**
DATE:	**27 February 2023**

Contents

1 NUMERICAL CALCULATION OF NAMEPLATE PARAMETERS 3

1.1 DERIVATION OF THE INDUCTION MOTOR EQUIVALENT CIRCUIT 3

2 VALIDATION OF OBTAINED PARAMETERS IN PLECS 4

2.1 LOCKED ROTOR TEST 4
2.2 NO LOAD TEST 5
2.3 FULL RATED LOAD 6

3 FURTHER APPROXIMATION OF THE PARAMETERS IN PLECS 9

4 NO LOAD AND FULL LOAD TEST AT LOW FREQUENCIES (HZ) 10

FIGURE 1: INDUCTION MACHINE IN THE LOCKED ROTOR STATE 4
FIGURE 2: INDUCTION MOTOR STARTING CURRENT 5
FIGURE 3: SYNCHRONOUS SPEED OF THE INDUCTION MOTOR 5
FIGURE 4: NO LOAD CURRENTS 6
FIGURE 5: RATED FULL LOAD SPEED 7
FIGURE 6: RATED MOTOR CURRENT 8
FIGURE 7: PULL-OUT TORQUE 8
FIGURE 8: TORQUE VS SLIP CHARACTERISTICS OF THE NEW EQUIVALENT MODEL 9
FIGURE 9: STARTING TORQUE OF THE NEW MODEL 10
FIGURE 10: REPRESENTATION OF THE MAGNETIC FLUX IN THE INDUCTION MOTOR 11
FIGURE 11: NO LOAD AND FULL LOAD OPERATIONS AT 5HZ 11
FIGURE 12:MAGNETIC FLUX OF THE INDUCTION MOTOR WITHOUT THE VOLTAGE BOOST 12
FIGURE 13: NO LOAD AND FULL LOAD OPERATIONS AT 5HZ WITHOUT THE VOLTAGE BOOST 12
FIGURE 14: NO LOAD AND FULL LOAD OPERATIONS AT 10HZ WITHOUT A VOLTAGE BOOST 13

Main Objectives

I. The main objective of this coursework is to correctly derive the different motor performance parameters from manufacturers' datasheet.
II. Be able to simulate in PLECS using the Direct Online method to do the no load test and locked rotor test to verify the derived nameplate parameters.
III. Analysing the performance of an induction motor at low frequencies using variable voltage variable frequency control

1 Numerical Calculation of Nameplate Parameters

1.1 Derivation of the Induction motor Equivalent Circuit

In the table below is the datasheet used to calculate the nameplate parameters of the Induction motor:

Table 1.1: Manufacturer Induction Motor Datasheet

Motor type	Rated power (kW)	Rated speed (rot/min)	Current (A)	η			cosφ	Is/In	Ts/Tn	Tmax/Tn	Mass (kg)
				100%	75%	50%					
ES-ASU 200La-6	18.5	977	34.2	90.5	90.5	89.5	0.87	7.7	2.3	3.3	257

- $S = \sqrt{3}V_{ll}I_n = \sqrt{3}\,(400)(34.2) = 23.694kVA$
- $Ns = 1000rpm\ since\ Nm\ is\ 977rpm$
- $s = \frac{Ns-Nm}{Ns} = \frac{1000-977}{1000} = 0.023\ slip\ at\ full\ load$
- $Pin = S\cos\varphi = 23.694kVA\ (0.87) = 20.614kW$
- $\eta = \frac{Pout}{Pin} = \frac{18.5kW}{20.614kW} = 0.897$

At full load slip is close to zero hence to get the shunt parameters (magnetizing reactance) of the equivalent circuit we do a short circuit test but at rated full load conditions. The following procedure was followed:

- $I_{no\ load} = I_N \sin\varphi = 34.2\ \sin 29.54° = 16.862A$
- $I'_R = I_N\ \cos\varphi = 34.2\ \cos 29.54° = 29.754A$
- $V_s = I'_R.\frac{R'_R}{s} \gg R'_R = \frac{s.V_s}{I'_R} = 0.023\left(\frac{\frac{400}{\sqrt{3}}}{29.754}\right) = 0.179\Omega$
- $V_S = jw_eL_mI_{no\ load} \gg L_m = \frac{\frac{400}{\sqrt{3}}}{2\pi 50Hz\ (16.862A)} = 0.044H$

To calculate the stator resistance the Power losses were calculated for different percentage values of load applied at the motor shaft.

- $Pin_{100\%} = 20614.18\ W \gg Pout_{100\%} = 0.905(20614.18W) = 18.66kW$
- $P_{loss-100\%} = Pin_{100\%} - Pout_{100\%} = 1.9542kW$
- $Pin_{50\%} = \sqrt{3}\,(400)(0.5\ I_N\ \cos\varphi = 14.877A) = 10307.09\ W$
- $Pout_{50\%} = 0.895(10307.09W) = 9224.844W$
- $P_{loss-50\%} = Pin_{50\%} - Pout_{50\%} = 1082.246W$

Using the value of rotor resistance referred to the stator (R'_R) above and then subtracting the 2 power losses we obtain:

- $(3.509 \times 10^3 - 1.517 \times 10^3)R_S + 474.13W - 118.85W = 1.9542kW - 1082.246W$

- $\gg R_S = \frac{516.674W}{1992A^2} = 0.26\Omega$

The starting of the motor has the same effect as doing the blocked rotor test. Hence, from the data obtained at the starting of the induction motor we can calculate the remaining series parameters (leakage inductances).

- $V_S = jw_e(l_S + l'_R)I_S \gg L_m = \frac{\frac{400}{\sqrt{3}}}{2\pi 50Hz\,(263.34A)} = 2.791mH$
- $\frac{l_S}{l_R} = \frac{R_S}{R'_R} = 1.453 \gg l_S = 1.453 l_R$
- $l'_R = 1.138mH,\ l_S = 1.654mH$

2 Validation of Obtained Parameters in PLECS

The parameters obtained the in previous selection were used to create a simulation model in PLECS simulation platform to validate the figures of merit listed in the data sheet (Table 1.1).

2.1 Locked Rotor Test

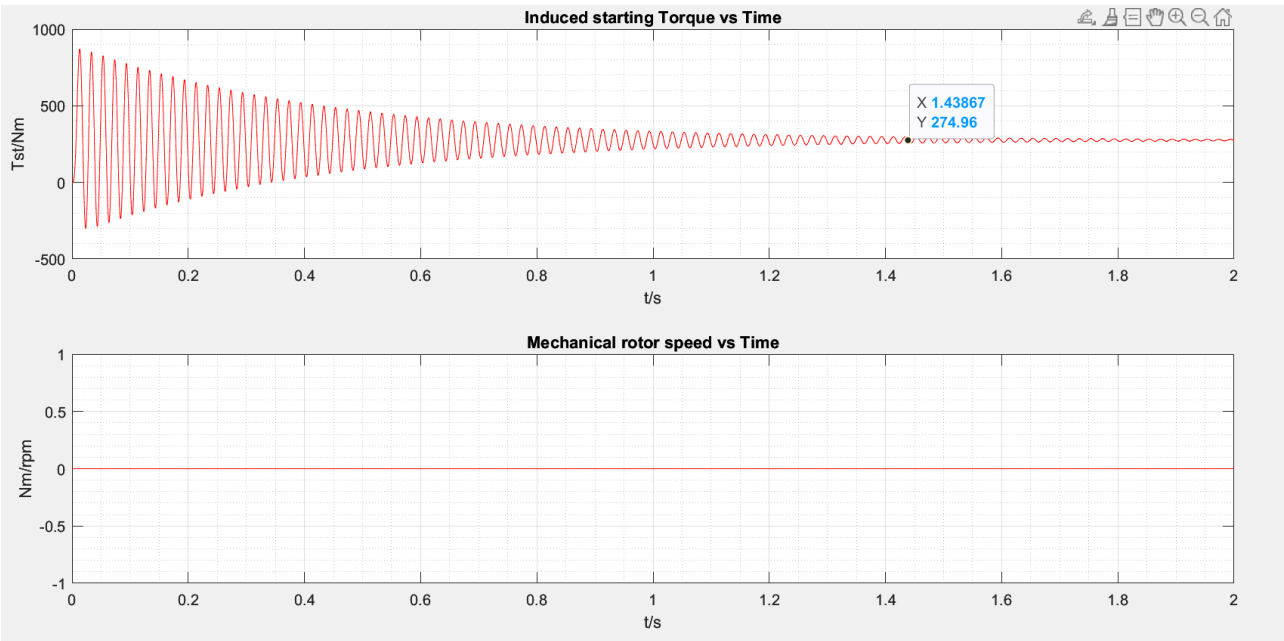

Figure 1: Induction Machine in the Locked rotor state

From the above it can be seen that mechanical rotor speed is at zero and therefore the induction machine is not rotating. The starting Torque was calculated and from the above graph it can be seen that is equal to 274.96Nm. When comparing this value with the one in the datasheet of Tst = 415.88Nm it can be seen that the error is about 33%. Although the equivalent circuit model parameters obtained in Section 1 were approximated, the error is too significant because there are limitations to the simulation model used in PLECS when calculating the induced starting torque using the DOL method. Some improvements to the equivalent circuit model parameters will be done in the later section to reduce this error.

In Figure 2.2 it can be seen that the starting current of the induction motor has a peak magnitude of 336.207A line current which if we change it to rms value is equivalent to 237.734A. From the datasheet the starting current is 7.7 times the rated current of 34.2A and therefore it is equal to 263.34A which is closer to the one obtained in the simulation. Since the rotor magnetic field is generated by the stator magnetic field through induction and for this to happen the flux generated by

the stator core need to pass through an air gap which is not a highly permeable material huge currents flow in the machine at the start.

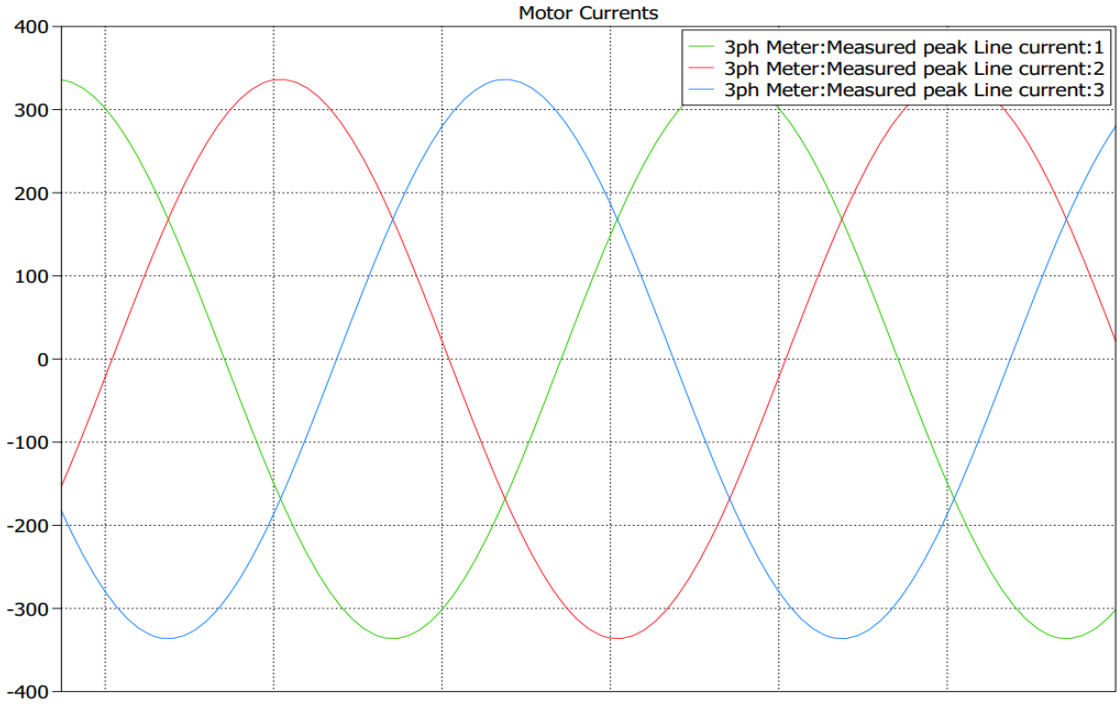

Figure 2: Induction Motor starting current

2.2 No Load Test

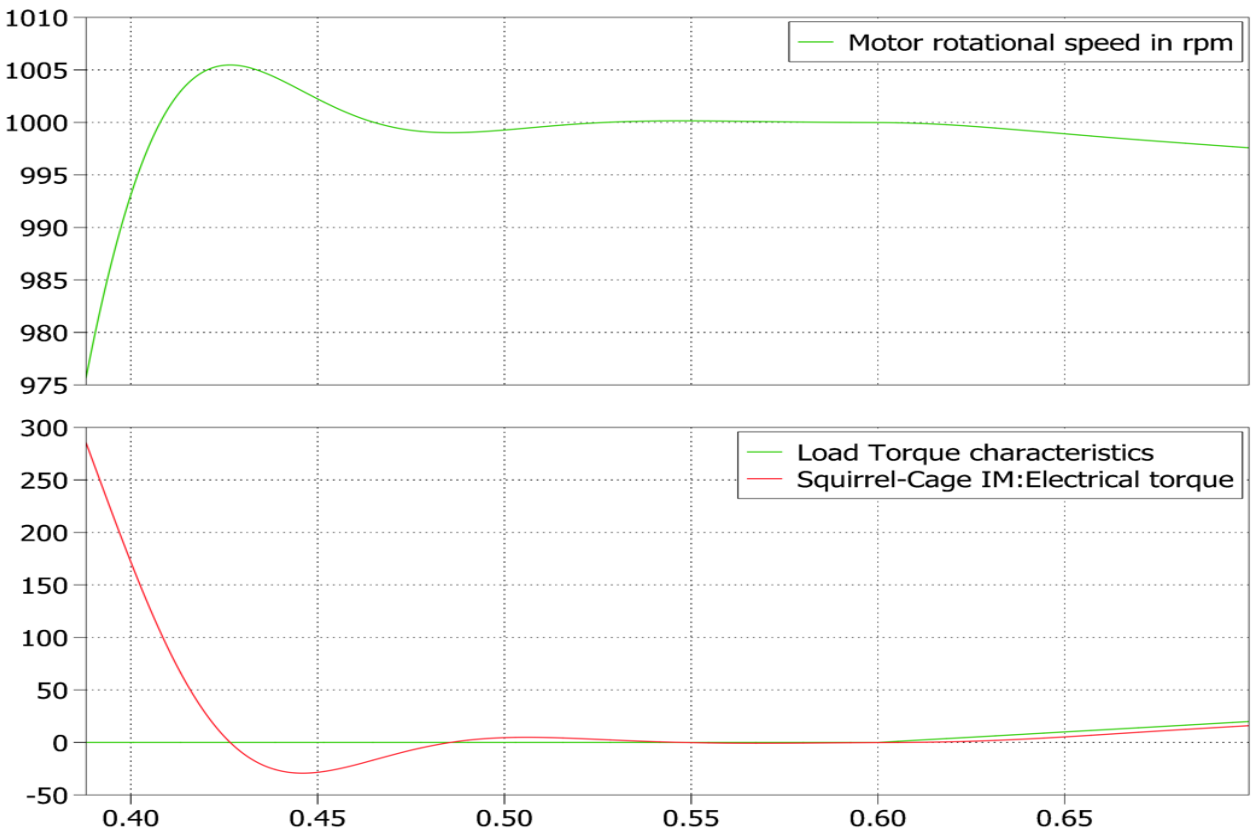

Figure 3: Synchronous Speed of the Induction Motor

In Fig 2.3 we can see that a Nm = 1000rpm the induced torque is zero and this is equivalent to the synchronous speed (Ns). This value is also equal to the one calculated in the first section which is the synchronous speed of a 6 pole induction machine.

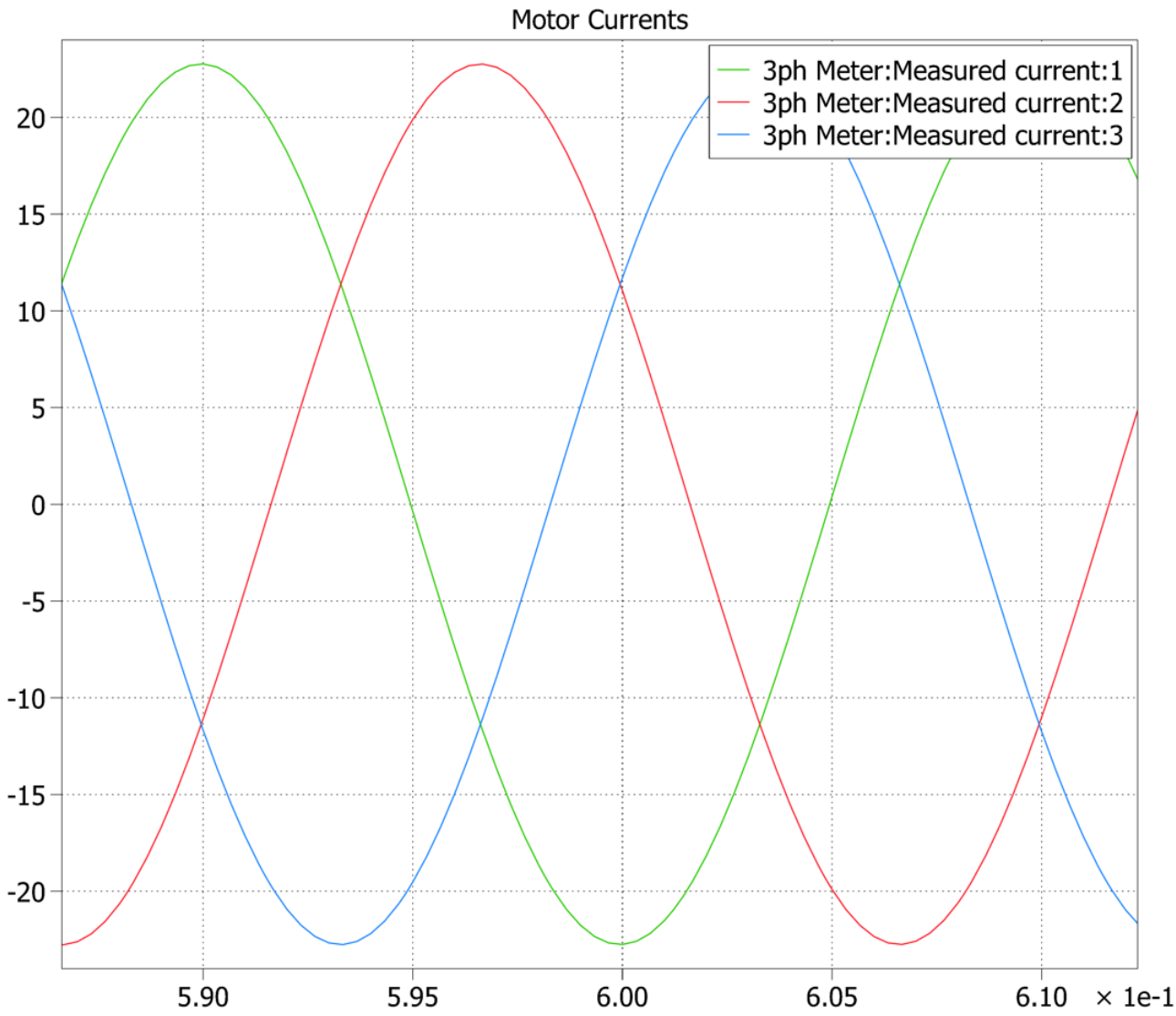

Figure 4: *No load Currents*

The no load current shown from the above graph is 22.7625A which in rms value is 16.096A and comparing this value with the calculated of 16.862A we see that the calculated value is almost the same as the value in the simulation model. It also shows that our approximation is smaller than the actual value.

2.3 Full rated load

Using the information provided in the datasheet we can derive the full load Tn (Torque at rated load) from the equation below:

- $T_N = \frac{18.5kW}{\frac{2\pi}{60}(977rpm)} = 180.82Nm$

The speed 977rpm in the datasheet corresponds to the actual speed of the motor when an 180.82Nm load torque is applied to it.

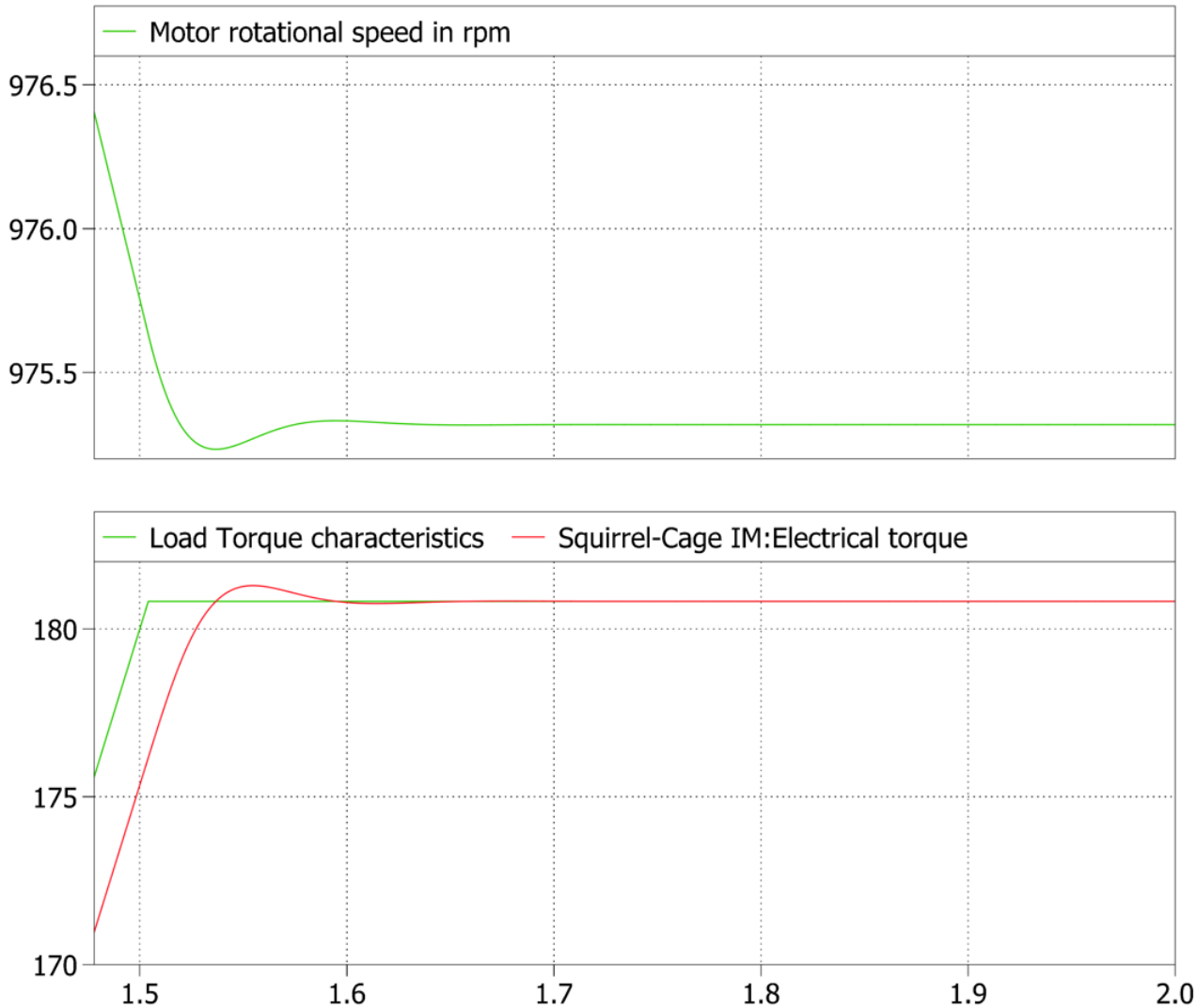

Figure 5: *Rated full load Speed*

The rated speed is about 975.332rpm is very close to 977rpm which shows that the approximation of the rotor resistance referred to the stator is very close to the true value. But, since the rotor resistance is directly proportional to the slip it can be shown that the approximation made is bigger than the true resistor value. Hence it should be slightly lowered for better results.

At full load it can be seen (Fig 2.6) that the full load current is 48.015A peak which means that the rms value is 33.95A. This is very close to the rated current of 34.2 A. Using both the full load current and the no load current we can get the power factor.

- $\sin\varphi = \frac{16.096}{33.95} \gg \varphi = 28.3° \gg \cos 28.3° = 0.88$
- This is close to the power factor of 0.87 given in the datasheet
- $I'_R = \sqrt{33.95^2 - 16.096^2} = 29.892A$
- $Pout = 3{I'_R}^2 R'_R \left(\frac{1-s}{s}\right) = 18946.13W \gg where\ s = \frac{1000-975.33}{1000} = 0.0247$
- Both the output power and the slip at full load are close to the specified values in the data
- $Pin = \sqrt{3}\,(400)(29.892) = 20709.785W$

- $\eta = \frac{Pout}{Pin} = 0.915$

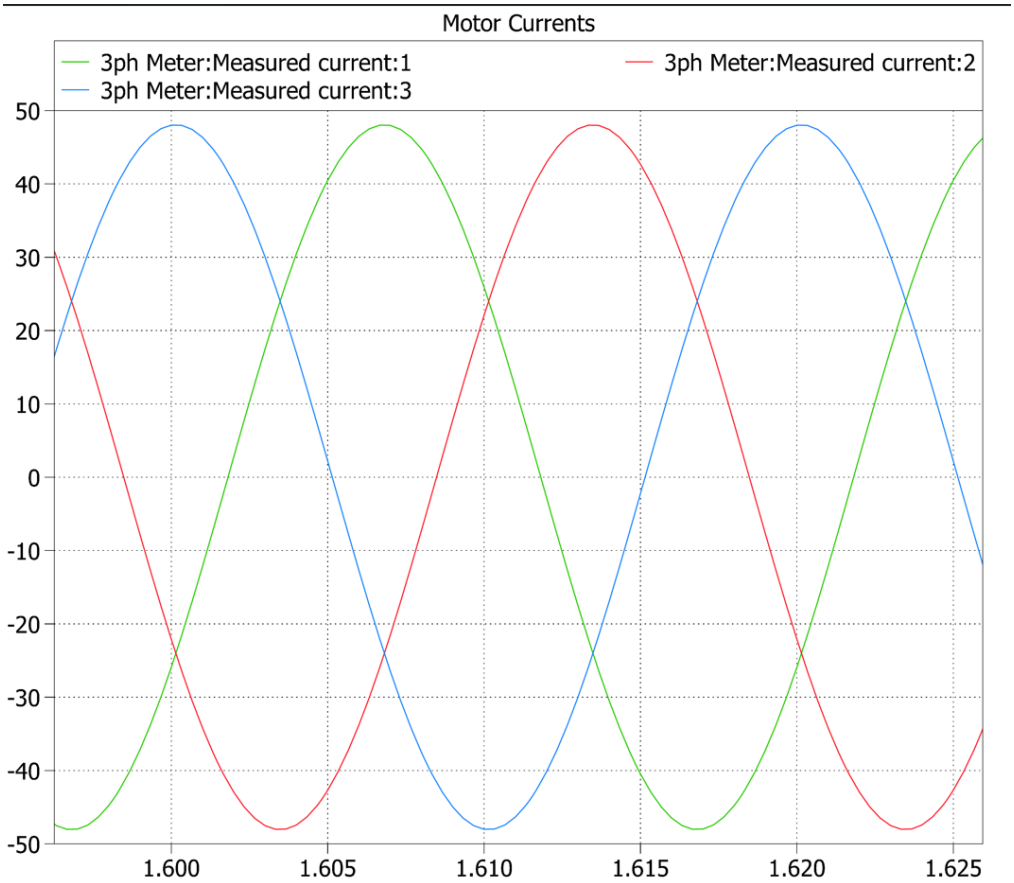

Figure 6: Rated motor current

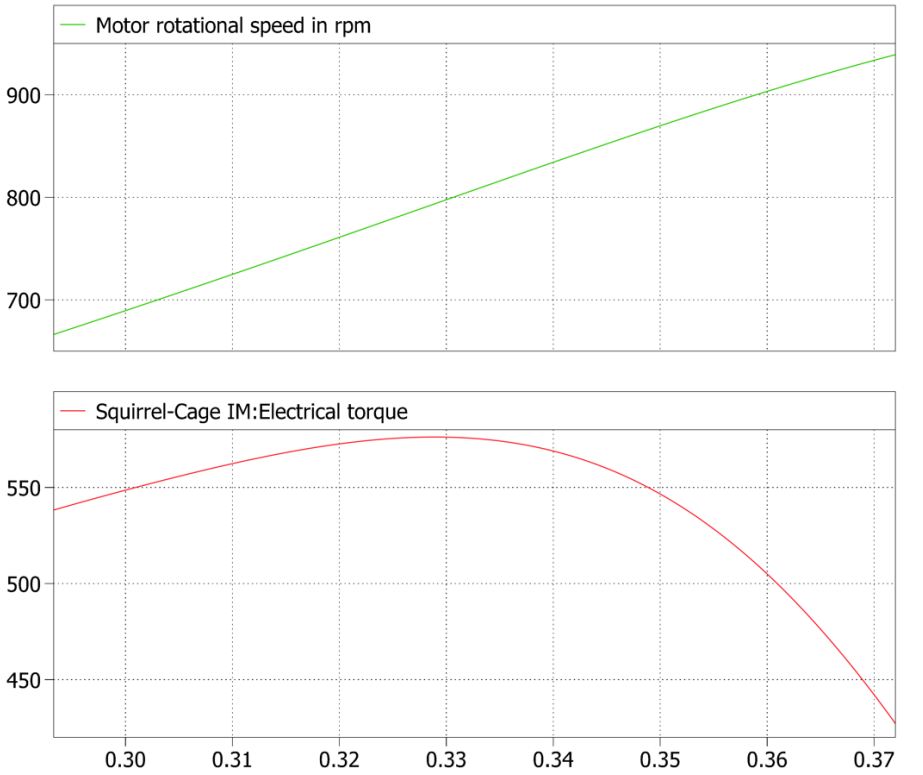

Figure 7: Pull-out Torque

The above figure shows that the Pull-out torque of the model is 576.209Nm. Comparing this with the one provided in the data of 596.708Nm it is very reasonable since both the magnetizing inductance and rotor resistance are slightly different from the true values.

3 Further Approximation of the Parameters in PLECS

Using the following analysis below new approximations were made.

- Slip is directly proportional to the rotor resistance. Hence as the rotor resistance increases and slip increases and the rotational speed decreases. Therefore, need to lower my rotor resistance to increase my current rotational speed.
- Changing rotor resistances does not affect the value of Pull-out torque.
- Maximum torque and starting torque are directly proportional to the square of the induced voltage in the rotor at s = 1 which is itself directly proportional to the square of the magnetizing inductance
- To increase both the value of the starting and maximum torque I need to increase my magnetizing inductance. This makes sense since the no load current measured in the simulation was lower than the one used in the calculations.
- Rotor leakage inductance is directly proportional to both starting and maximum torque.

The parameters which were changed, and their new values are specified below:

1) $R'_R = 0.16536\Omega$
2) $l'_R = 1mH$
3) $L_m = 45mH$

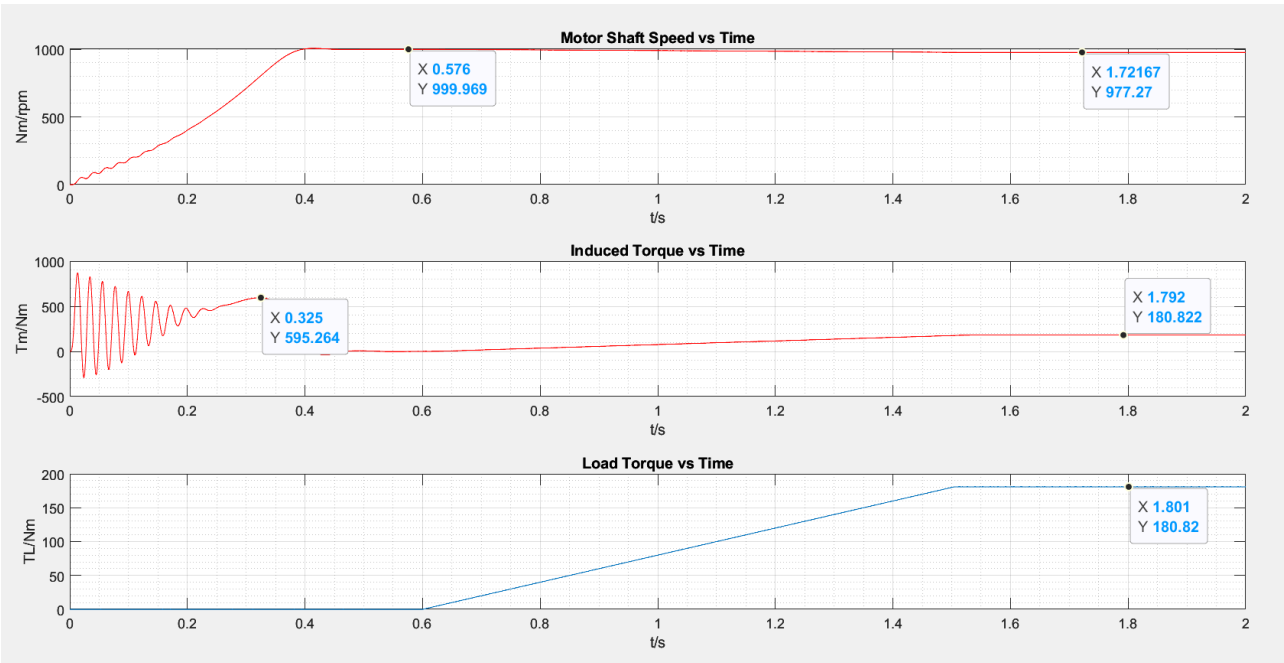

Figure 8: Torque vs slip characteristics of the new equivalent model

It can be seen that the maximum torque is now 595.264Nm which is much closer to the one in the data. The same also can be said for the rotational speed which is now exactly 977rpm at rated load.

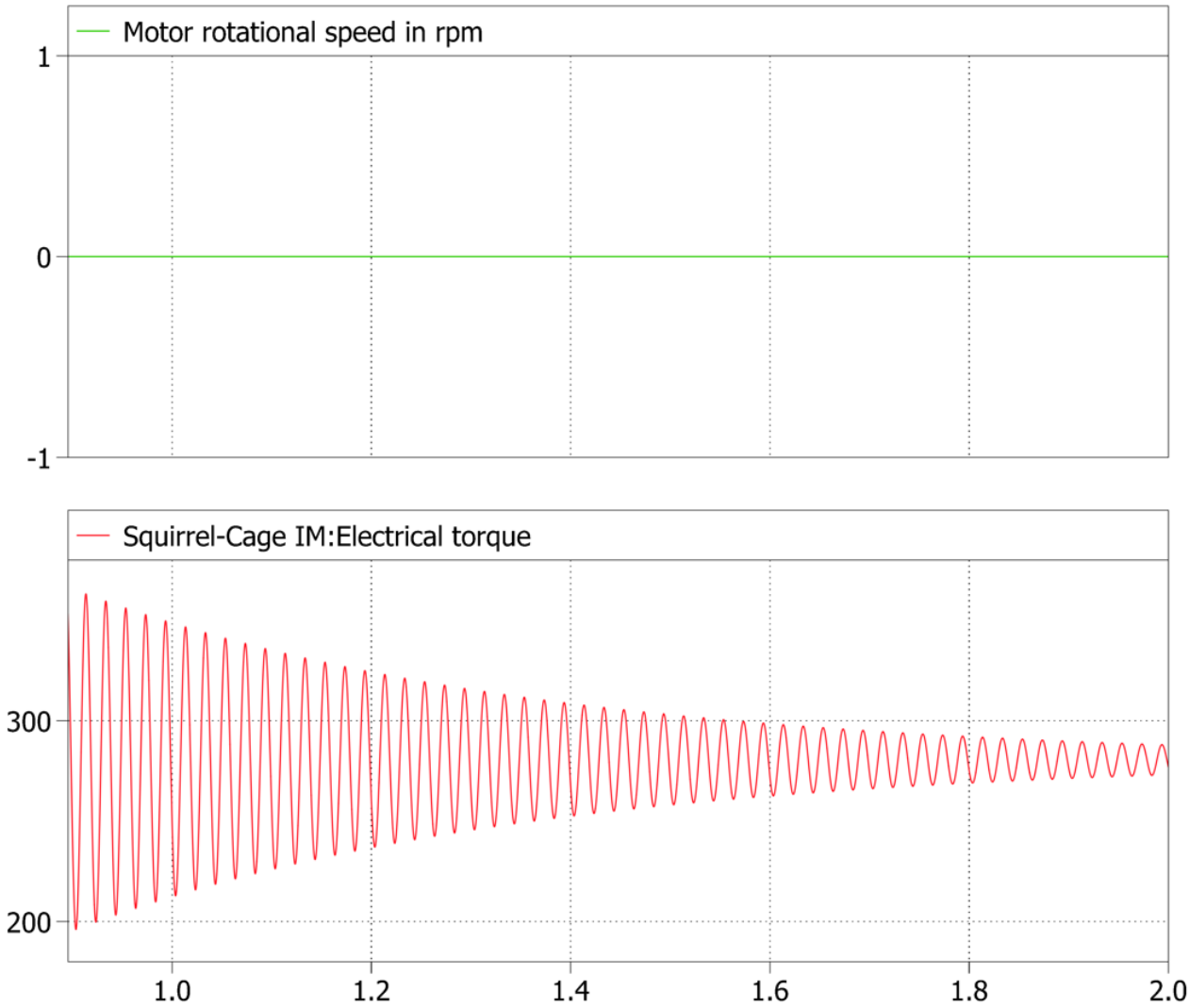

Figure 9: Starting Torque of the new model

The starting torque is now 281.037Nm which has now an error of $\frac{415.88 - 281.037}{415.88} = 32\%$

4 No Load and Full Load test at Low frequencies (Hz)

- The figure below shows one of the draw backs of scalar control it can be seen that the voltage boost is too large and hence the magnetic core has reached saturation, and this can damage the machine. On the other hand, if we supply a very low voltage boost the current will not reach the no load current required to build the magnetic field needed for the normal operation of the machine.
- From the figure below it can also be seen that once the machine is loaded the magnetic flux drops to the no load magnetic flux and the magnetic core is no longer operating in saturation state.

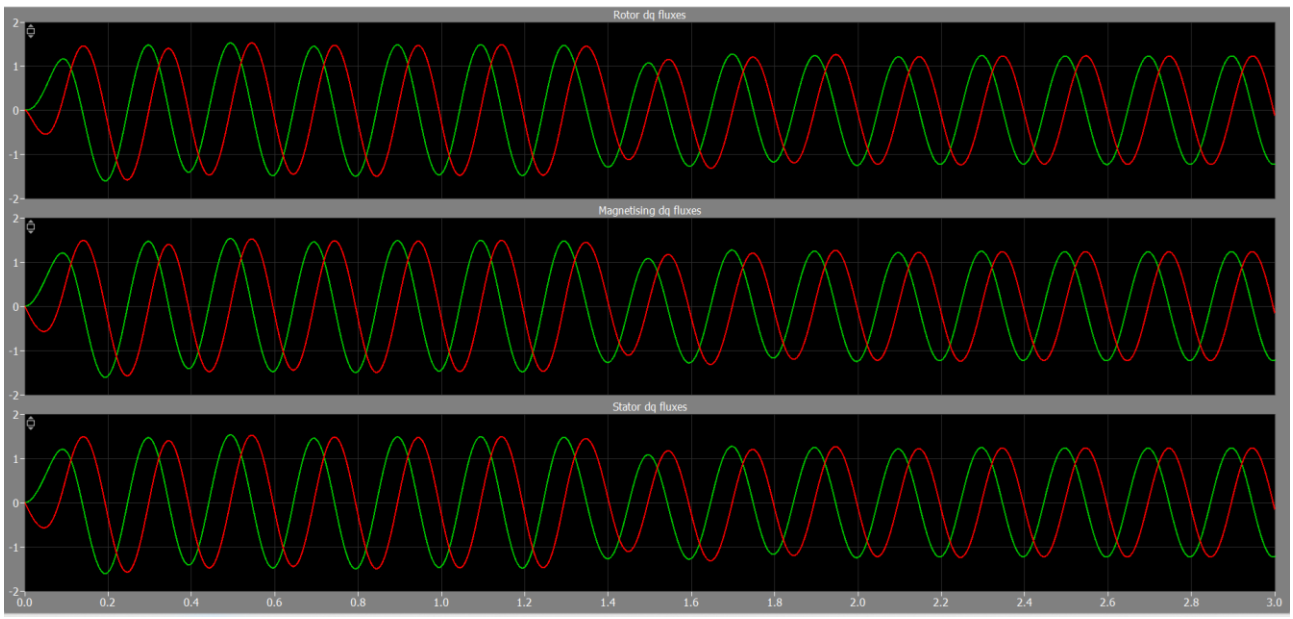

Figure 10: Representation of the magnetic flux in the Induction motor

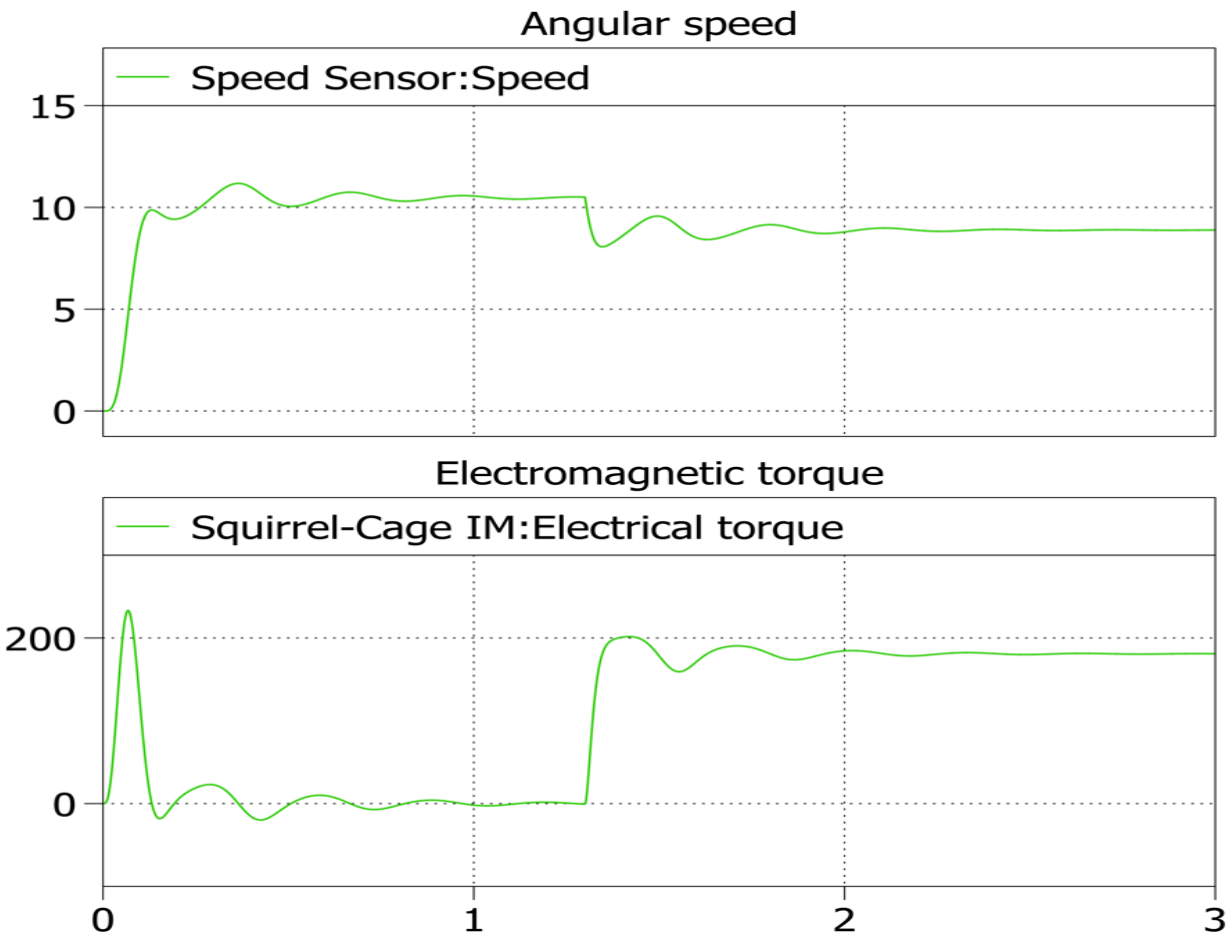

Figure 11: No load and Full load operations at 5Hz

The no load synchronous speed is 10rpm and once loaded the rotational speed drops down to 8rpm which is a 20% drop from the synchronous speed this is not something which happens for high frequencies. The induction motor is still able to rotate the rated load therefore speed control is still possible at low frequency because we only need a minimum of 1.15Hz (slip at rated load) to rotate the rotor when fully loaded.

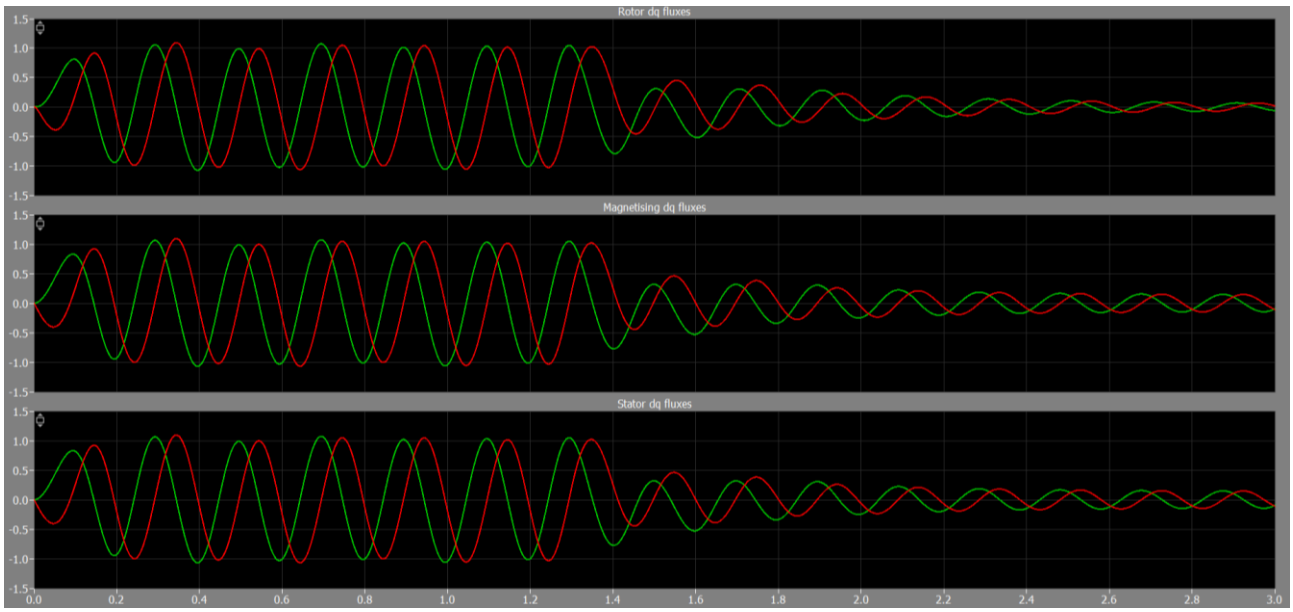

Figure 12:Magnetic flux of the induction motor without the voltage boost

At no load the induction motor generates the required magnetic field to run the motor at low speed but at full load the magnetic flux weakens greatly and goes beyond 0.5Wb. This is one other disadvantage of operating at low frequency that when loaded the induction machine magnetic field strength drops down significantly.

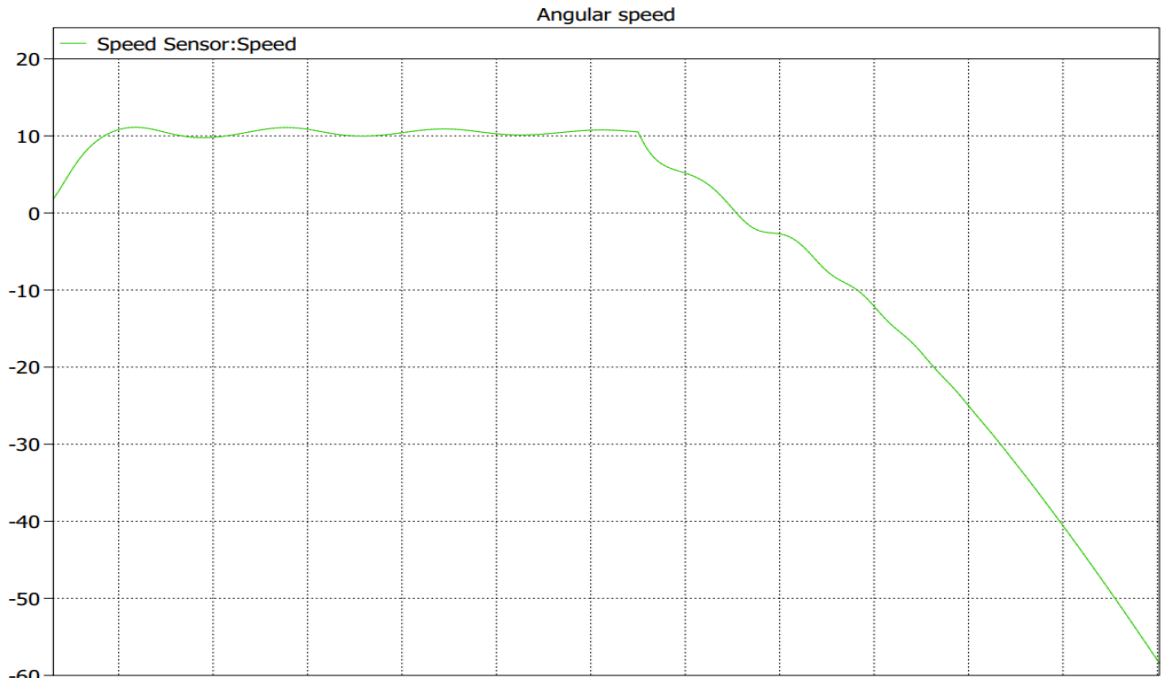

Figure 13: No load and Full load operations at 5Hz without the voltage boost

At full load the motor will rotate in the opposite direction of the synchronous speed (magnetic field) until it reaches negative speeds (generating mode).

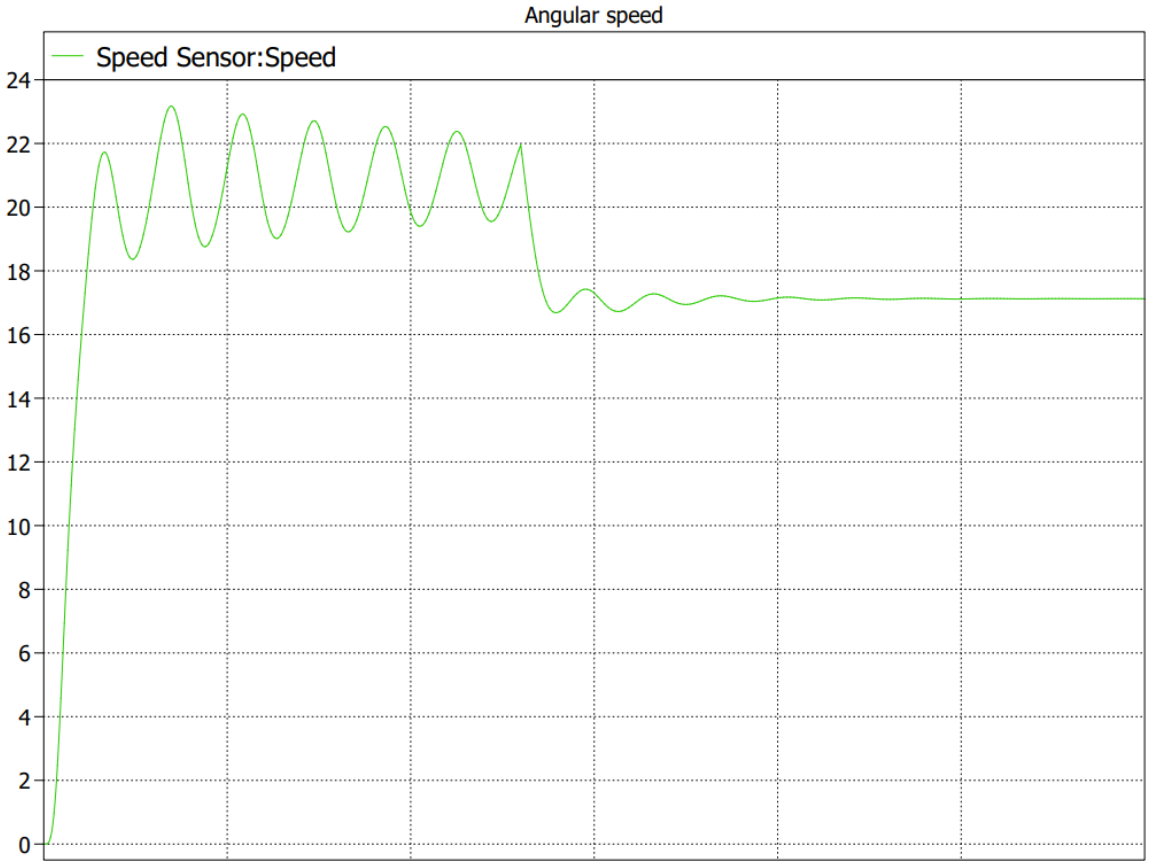

Figure 14: No Load and Full Load operations at 10Hz without a voltage boost

It can be noted that at 10Hz frequency the induction motor operates smoothly for both full Load and no Load conditions. The ratio of increase in frequency is not directly proportional to the ratio of increase in speed.